SUMMARY

Minerals are inorganic substances that play major roles in many metabolic functions. They are divided among two classes, macro elements and micro elements. In general, most of these elements are components of enzymes, which are catalysts of chemical reactions in the body. Additionally, minerals regulate and control the normal function of human and animal tissues, muscles, and organs.

Macro elements including sodium, potassium, calcium, magnesium, and phosphorus in large amounts. In specific, sodium, chloride and potassium play a vital role in maintaining proper pH balance of blood and water. Phosphorous & calcium acts as a major structural component of bones and teeth. These minerals are typically available in dairy products, green vegetables, nut, beans, milk, egg, meat, fish and table salt. Deficiency of macro elements like sodium, calcium, calcium, magnesium and potassium causes diarrhoea, dehydration, osteoporosis, tetany and muscle weakness respectively.

Micro elements also called trace elements are needed in small amounts but important for the proper functioning of the body. Some of the essential trace minerals include iron, zinc, copper, selenium, chromium and iodine. Iron carries oxygen throughout the body in the blood. Zinc helps in growth, reproduction and wound healing. Selenium maintains the fertility. Chromium accelerates the utilization of glucose and reduces serum cholesterol. Iodine responsible for the synthesis of thyroid hormones and regulates the energy metabolism. Drinking water, vegetables, whole grains, legumes, cereals, nuts, seafood, liver, kidney, meat, & fish are the rich sources of micro elements. Iodine deficiency leads to simple goiter and iron deficiency causes anaemia and diarrhoea. Shortage of micro elements leads to loss of weight, dental caries, skeletal abnormalities, weight loss and disturbance in metabolism are the other effects.

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